

FOURIER TRANSFORM, FOURIER SERIES, AND THEIR APPLICATIONS

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ABSTRACT. The project is focused on the rigorous study of Fourier analysis, primarily from two books [1] and [2]. In particular, we study two important applications of Fourier analysis, one leads to the mathematical proof of the Heisenberg Uncertainty Principle, while the other one is concerned with the Radon transform, and its properties. To that end, we study all the relevant definitions, prove basic and advanced properties, and theorems from Chapters 2,3, 5, and 6 of [1] relevant to the two above applications.

1. History & Motivation

Fourier analysis was first developed to solve the physical problems of modeling a vibrating string and heat flow. To that end, one may apply relevant physical laws to derive the heat and wave partial differential equations, yet it remains to solve both partial differential equations. This section explores how the genesis of Fourier analysis began from solving the heat and wave equations.

We begin by discussing the vibrating string problem, observing a phenomenon best understood in the context of music. Musicians have known since at least Pythagoras that the sound produced by plucking a string in a musical instrument is made up of several resonant frequencies. These frequencies include the fundamental frequency, i.e. pure tones, and its harmonics, i.e. overtones. The individual timbre of the instrument is determined by its harmonics, which is why one can play the same note on a guitar and a piano and still hear distinct “colors” of sounds.

We can implement the linearity of the wave equation to model the phenomenon of overtones. Suppose we know one solution of the wave equation which corresponds with a pure tone made up of a single frequency. Then using superposition, a complex tone composed of several harmonics such as one from a piano would correspond to a linear combination of simpler solutions. This implies the now well known fact that linear combinations of solutions to the wave equation are also solutions to the wave equation.

Fourier then abstracted this idea of decomposing periodic functions into linear combinations of sinusoidal waves at different frequencies, i.e. Fourier series, which would begin the subject of Fourier analysis. Stein explains, “Joseph Fourier (1768 - 1830) was the first to believe than an ‘arbitrary’ function F could be given as a series (1.1). In other words, his idea was any function is the linear combination (possibly infinite) of the most basic trigonometric functions $\sin mx$ and $\cos mx$, where m ranges over the integers. Although this idea was implicit in earlier work, Fourier had the conviction that his predecessors lacked, and he used it in his study of heat diffusion; this began the subject of “Fourier analysis.” This discipline, which was first developed to solve certain physical problems, has proved to have many applications in mathematics and other fields as well, as we shall see later.” The equation referenced in this quote is:

$$(1.1) \quad F(x) = \sum_{m=-\infty}^{\infty} a_m e^{imx}, \quad \text{with} \quad a_m = \frac{1}{L} \int_a^b f(x) e^{-2\pi imx/L} dx.$$

This report is organized as follows. In section 3, we define Fourier series and its properties. In section 4, we prove two important convergence theorems of Fourier series: the Mean-square and Pointwise convergence theorems. The theory of Fourier series however only applies to periodic functions on the circle. In section 5, we shall extend this theory to apply to non-periodic functions through the Fourier transform as a somewhat “continuous” analogue of Fourier series. In our final section 6, we prove our two main results:

the Heisenberg Uncertainty Principle and the Uniqueness and Reconstruction theorems of the Radon transform.

2. Preliminaries & Notations

Before defining Fourier series, we introduce notation used throughout this paper.

Define an function f to be **even** on one of its periods $[a, b]$ if for all $x \in \left[\frac{b-a}{2}, -\frac{b-a}{2}\right]$, the following condition is true:

$$f\left(\frac{a+b}{2} + x\right) = f\left(\frac{a+b}{2} - x\right).$$

Similarly define a function which is odd on one of its periods. Throughout the paper we will refer to these functions as simply even or odd.

Definition 2.1. Define the space of all square integrable functions on $[-\pi, \pi]$ as:

$$L^2([-\pi, \pi]) = \left\{ f : \int_{-\pi}^{\pi} |f(t)|^2 dt < \infty \right\}.$$

When discussing orthogonal bases of $L^2([-\pi, \pi])$, we denote by $\{e_n = e^{inx}\}$ our sequence of basis functions. Each inner product (and hence each norm as well) used will be between two Riemann integrable functions on $[0, 2\pi]$, $f, g \in L^2([-\pi, \pi])$. We have the usual inner-product on $L^2([-\pi, \pi])$ as:

$$(f, g) = \frac{1}{2\pi} \int_0^{2\pi} f(x) \overline{g(x)} dx.$$

We define two classes of functions used throughout the paper:

Definition 2.2. A function f defined on $\mathbb{R}$ is said to be of **moderate decrease**, denoted $f \in \mathcal{M}(\mathbb{R})$, if f is continuous and $\exists A > 0$ such that:

$$|f(x)| \leq \frac{A}{1 + |x|^{1+\epsilon}} \quad \forall x \in \mathbb{R}, \quad \forall \epsilon > 0.$$

We use this class of functions because of the special property that any function of moderate decrease can be integrated on $\mathbb{R}^d$ (see appendix 3.1 of [1]).

Definition 2.3. The **Schwartz space** on $\mathbb{R}$, denoted $\mathcal{S} = \mathcal{S}(\mathbb{R})$ consists of the set of all indefinitely differential functions f so that f and all its derivatives $f', f'', \dots, f^{(l)}, \dots$, are **rapidly decreasing** in the sense that:

$$\sup_{x \in \mathbb{R}} |x|^k |f^{(l)}(x)| < \infty \quad \forall k, l \geq 0.$$

In other words, $f(x)$ and all its derivatives decay faster than any polynomial can grow in the sense that their product is bounded and will approach zero as $x \rightarrow \infty$. Throughout this paper, the term “Schwartz function” will be used interchangeably with the term “rapidly decreasing function”. Any function that is rapidly decreasing will also be of moderate decrease, so $\mathcal{S}(\mathbb{R}) \subset \mathcal{M}(\mathbb{R})$. For example, $f_2(x) = e^{-\pi x^2}$ decreases rapidly. However, $f_1(x)$ is not Schwartz since we may multiply f_1 by the polynomial x^3 and get an unbounded function.

We define the convolution operation, which among other applications, will be central to proving the Fourier inversion formula.

Definition 2.4. If $f, g \in \mathcal{S}(\mathbb{R})$, then the **convolution** of f and g is defined to be:

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt.$$

When we begin using partial derivatives after defining the multi-dimensional Fourier transform in section 5.4, we will use the notation $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d), \beta = (\beta_1, \beta_2, \dots, \beta_d)$ for multi-indexes, i.e. d-tuples of non-negative integers such that $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_d^{\alpha_d}$.

We will need the following notation for surface elements $d\sigma(\gamma)$ to perform a plane integral in section 6.2. Frequently used in our study, the spherical coordinate-change is given by:

$$\begin{cases} x_1 &= r \sin \theta \cos \phi \\ x_2 &= r \sin \theta \sin \phi \\ x_3 &= r \cos \theta, \end{cases}$$

where $0 < r$, $0 \leq \theta \leq \pi$, and $0 \leq \phi \leq 2\pi$. Since the Jacobian of the change of variables to these spherical coordinates is $r^2 \sin \theta$ such that:

$$\int_{\mathbb{R}^3} f(x) dx = \int_0^{2\pi} \int_0^\pi \int_0^\infty f(r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta) r^2 dr \sin \theta d\theta d\phi.$$

If g is a function on the unit sphere $S^2 = \{x \in \mathbb{R}^3 : |x| = 1\}$, and $\gamma = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, we define the surface element $d\sigma(\gamma)$ by:

$$\int_{S^2} g(\gamma) d\sigma(\gamma) = \int_0^{2\pi} \int_0^\pi g(\gamma) \sin \theta d\theta d\phi.$$

Therefore, we find the following equivalent method to integrate across all of $\mathbb{R}^3$:

$$\int_{\mathbb{R}^3} f(x) dx = \int_{S^2} \int_{\mathbb{R}} f(r\gamma) r^2 dr d\sigma(\gamma).$$

Finally, we will sometimes decompose integrals over $\mathbb{R}$:

$$\int_{\mathbb{R}} f(x) dx = \text{“head”} : \int_{|x| \leq N} f(x) dx + \text{“tail”} : \int_{|x| > N} f(x) dx$$

The “head”, as we will refer to it, is the first integral over a bounded domain for some $N \in \mathbb{N}$. The “tail” is the integral over the unbounded domain not included in the head.

3. Properties of Fourier Series

In this section we provide the proof for some fundamental properties of the Fourier series, starting from the basic definition.

Definition 3.1. If f is integrable on $[a, b]$, then the **Fourier series** of f is given formally by ($n \in \mathbb{Z}$)

$$f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x / L}, \quad \hat{f}(n) = \frac{1}{L} \int_a^b f(x) e^{-2\pi i n x / L} dx,$$

where $\hat{f}(n)$ are defined to be the **Fourier coefficients** of f , and $L = b - a$.

Whenever f is defined on the circle, then $L = 2\pi$. It is usually preferable to use a symmetrical domain, so often we let $a = -\pi$ and $b = \pi$, although any $a, b \in \mathbb{R}$ such that $b - a = 2\pi$ can be chosen.

The family of Fourier series is a special case of the more general family called **trigonometric series**, defined to be expressions of the form $\sum_{n \in \mathbb{N}} c_n e^{2\pi i n x / L}$ where $c_n \in \mathbb{C}$. Fourier series are a special case in that our coefficients $c_n = \hat{f}(n)$ are chosen to be Fourier coefficients. If a trigonometric series involves only finitely many non-zero terms, it is called a **trigonometric polynomial**. Its degree is the largest $|n|$ such that $c_n \neq 0$. Thus similarly, the N^{th} **partial sum** of the Fourier series of f given below ($N \in \mathbb{N}$) is one example of a trigonometric polynomial:

$$(3.1) \quad S_N(f)(x) = \sum_{|n| \leq N} \hat{f}(n) e^{2\pi i n x / L}.$$

Partial sums of Fourier series will be used often in section 4 when discussing the convergence of Fourier series.

We may obtain an equivalent definition of Fourier series by expanding our linear combination of complex exponential functions, $e^{2\pi i n x/L}$, to a linear combination of sines and cosines using Euler's formula. Let $\hat{f}_e(n) = [\hat{f}(n) + \hat{f}(-n)]$ and let $\hat{f}_o(n) = [\hat{f}(n) - \hat{f}(-n)]$. Then:

$$\begin{aligned} f(x) &\sim \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x/L} \\ &= \hat{f}(0) + \sum_{n \geq 1} \hat{f}(n) e^{2\pi i n x/L} + \hat{f}(-n) e^{-2\pi i n x/L} \\ &= \hat{f}(0) + \sum_{n \geq 1} \hat{f}(n) \left[\cos\left(\frac{2\pi n x}{L}\right) + i \sin\left(\frac{2\pi n x}{L}\right) \right] + \hat{f}(-n) \left[\cos\left(\frac{2\pi n x}{L}\right) - i \sin\left(\frac{2\pi n x}{L}\right) \right]. \end{aligned}$$

Hence by rearranging the last equation and substituting in $\hat{f}_e(n)$, $\hat{f}_o(n)$, we obtain our equivalent definition of Fourier series:

$$f(x) \sim \hat{f}(0) + \sum_{n \geq 1} \hat{f}_e(n) \cos\left(\frac{2\pi n x}{L}\right) + i \hat{f}_o(n) \sin\left(\frac{2\pi n x}{L}\right).$$

We now discuss some relevant properties of Fourier series. If a function f is even, then $\forall n \in \mathbb{N}$, $\hat{f}_o(n) = 0$. Therefore the only non-zero terms of the series are cosines. Furthermore:

$$\hat{f}_e(n) = \hat{f}(n) + \hat{f}(-n) = \frac{1}{L} \int_a^b f(x) \left(e^{2\pi i n x/L} + e^{-2\pi i n x/L} \right) dx = \frac{2}{L} \int_a^b f(x) \cos\left(\frac{2\pi n x}{L}\right) dx.$$

Similarly, if f is odd on $[a, b]$, then $\forall n \in \mathbb{N}$, $\hat{f}_e(n) = 0$, the series is composed entirely of sines, and:

$$\hat{f}_o(n) = \hat{f}(n) - \hat{f}(-n) = \frac{i}{L} \int_a^b f(x) \left(e^{2\pi i n x/L} - e^{-2\pi i n x/L} \right) dx = \frac{2i}{L} \int_a^b f(x) \sin\left(\frac{2\pi n x}{L}\right) dx.$$

Another property is that

$$\hat{f}(0) = \frac{\hat{f}_e(0)}{2} = \frac{1}{L} \int_a^b f(x) dx$$

is the average value of f over a single period L . Therefore, if f is odd, then $\hat{f}(0) = 0$.

One useful example of a Fourier series is that of the absolute value function on $[-\pi, \pi]$:

$$(3.2) \quad f(x) = |x| \quad \sim \quad \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}.$$

Notice that since our function is even on $[a, b]$, then only $\hat{f}_e(n)$ coefficients are non-zero, hence our series is made up entirely of cosines.

To motivate why to use trigonometric polynomials to approximate periodic functions, we see the following application (Exercise 2.6 from [1]) of finding some infinite sums we otherwise could not:

$$\sum_{n \text{ odd} \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Substitute $x = 0$ into (3.2) for the first identity:

$$f(0) = 0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \implies \frac{\pi^2}{8} = \sum_{n \text{ odd} \geq 1} \frac{1}{n^2}.$$

For the second inequality we use that:

$$\sum_{n \text{ even} \geq 1} \frac{1}{n^2} = \sum_{n \geq 1} \frac{1}{(2n)^2} = \frac{1}{4} \sum_{n \geq 1} \frac{1}{n^2} \quad \text{and} \quad \frac{1}{4} \sum_{n \geq 1} \frac{1}{n^2} + \frac{\pi^2}{8} = \sum_{n \geq 1} \frac{1}{n^2}.$$

As in the previous application, once we consider the convergence of Fourier series to their original function, due to the uniqueness of limits, it is natural to question if given a function corresponds to one unique Fourier series. This question is addressed by the following theorem.

Theorem 3.2 (Uniqueness of Fourier Series). *Suppose that f is an integrable function on the circle with $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$. Then $f(x_0) = 0$ whenever f is continuous at the point x_0 .*

Proof : Without loss of generality, let f be defined on $[-\pi, \pi]$, and argue by contradiction. We seek to contradict the assumption that $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$. We assume without loss of generality $f(0) > 0$, and select $B \in \mathbb{R}$ s.t. $\forall x, |f(x)| \leq B$. We will achieve the contradiction by building a sequence of trigonometric polynomials $\{p_k\}$ s.t. $0 < \hat{f}(k) = \int_{-\pi}^{\pi} p_k(x)f(x)dx \rightarrow \infty$ as $k \rightarrow \infty$.

Given any fixed $\epsilon > 0$, let our trigonometric polynomial be $p(x) = \epsilon + \cos(x)$. Define $p_k(x) = p^k(x)$ since we desire that the limiting function $\lim_{k \rightarrow \infty} p_k(x)$ approaches ∞ if $x = 0$ and approaches 0 otherwise. We partition our domain into three parts and integrate each separately. First the tail should vanish, i.e. if we let $I_1(k) = \int_{\delta \leq |x|} f(x)p_k(x)dx$, then $I_1(k) \rightarrow 0$ as $k \rightarrow \infty$. Second, there is a small portion of the domain between the head and tail that need only be non-negative, i.e. if we let $I_2(k) = \int_{\eta \leq |x| < \delta} f(x)p_k(x)dx$, then $I_2(k) \geq 0$ as $k \rightarrow \infty$. Finally the head must diverge, i.e. if we let $I_3(k) = \int_{|x| < \eta} p_k(x)f(x)dx$, then $I_3(k) \rightarrow \infty$ as $k \rightarrow \infty$.

Therefore we must choose η, δ subject to the following conditions. First, η, δ must also satisfy $|p(x)| < 1 - \epsilon/2$ whenever $\delta \leq |x| \leq \pi$. Then for x in the tail:

$$|p(x)| < 1 \implies \lim_{k \rightarrow \infty} |p_k(x)| = 0 \implies \lim_{k \rightarrow \infty} I_1(k) \leq \lim_{k \rightarrow \infty} 2\pi B(1 - \epsilon/2)^k = 0.$$

Second, choose η, δ s.t. $0 < f(0)/2 < f(x)$ and $0 \leq p(x)$ whenever $|x| < \delta \leq \pi/2$. Then $I_2(k) \geq 0$. Finally, choose η, δ s.t. $p(x) \geq 1 + \epsilon/2$ whenever $|x| < \eta$. Then for x in the head:

$$p(x) \geq 1 \implies \lim_{k \rightarrow \infty} p_k(x) = \infty \implies \lim_{k \rightarrow \infty} I_3(k) \geq \lim_{k \rightarrow \infty} 2\eta \frac{f(0)}{2} (1 + \epsilon/2)^k = \infty.$$

Therefore with our choice of η, δ , $\int_{-\pi}^{\pi} p_k(x)f(x)dx \rightarrow \infty$ as $k \rightarrow \infty$, so we achieve our desired contradiction. This concludes the proof when f is real-valued. In general, write $f(x) = u(x) + iv(x)$, where u, v are real-valued. If we define $\bar{f}(x) = \overline{f(x)}$, then:

$$u(x) = \frac{f(x) + \bar{f}(x)}{2} \quad \text{and} \quad v(x) = \frac{f(x) - \bar{f}(x)}{2i}.$$

Since $\hat{\bar{f}}(n) = \overline{\hat{f}(-n)}$, then the Fourier coefficients of u and v all vanish, hence $f = 0$ at its points of continuity. $\square$

Uniqueness follows immediately by assuming by contradiction that there exist two distinct integrable functions f and g whose Fourier series are equal, and applying theorem 3.2 to $f(x) - g(x)$, achieving the contradiction that $f(x) = g(x) \forall x$. If $\theta_0 \neq 0$, then appropriately translate our chosen δ and η , keeping the relative distances between $\pm\delta$, $\pm\eta$, and θ_0 the same. Then we build our trigonometric polynomial to peak at θ_0 instead of 0. Otherwise, the proof is the same.

4. Convergence of Fourier Series

4.1. Best Approximation.

To begin, we prove some results to be used in the proof of the mean-square convergence theorem. Recall from linear algebra orthonormal bases. Observe that

$$\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n \in \mathbb{Z}}$$

is an orthonormal basis of $L^2([-\pi, \pi])$. We introduce the notation $\{e_n = e^{inx}\}$. Hence we may approximate any function $f \in L^2([-\pi, \pi])$ with a linear combination of our basis functions. With more basis functions used in the linear combination, our approximation is improved in the sense that certain error is reduced. Therefore, by letting our number of basis functions used, N , approach infinity, our approximation can become arbitrarily accurate. It is by taking this limit that one would formally obtain the Fourier series of

our original function.

However, if we use $S_N(f)$ as our approximation, then our projection space is $2N + 1$ dimensional. Even if f is not in the span of our finite basis, we can show $S_N(f)$ is the projection of f onto the span of our basis, and hence will be a better approximation than any other linear combination of the same basis functions. This is the idea of the best approximation lemma, which roughly states that the partial sum of f 's Fourier series will be the best possible approximation of f using our basis.

Before proving the best approximation lemma, we must prove a lemma which describes how our error function, the difference between f and its approximation $S_N(f)$, is orthogonal to the span of our basis functions.

Lemma 4.1. *If f is an integrable function on the circle with S_N as the N^{th} partial sum of Fourier coefficients of f , then $\forall b_n \in \mathbb{C}$:*

$$(f - S_N) \perp \sum_{|n| \leq N} b_n e_n.$$

Proof : Notice the important property that given an integrable function on the circle f , each of f 's Fourier coefficients a_n are given by the inner product of f with the n^{th} basis function e_n :

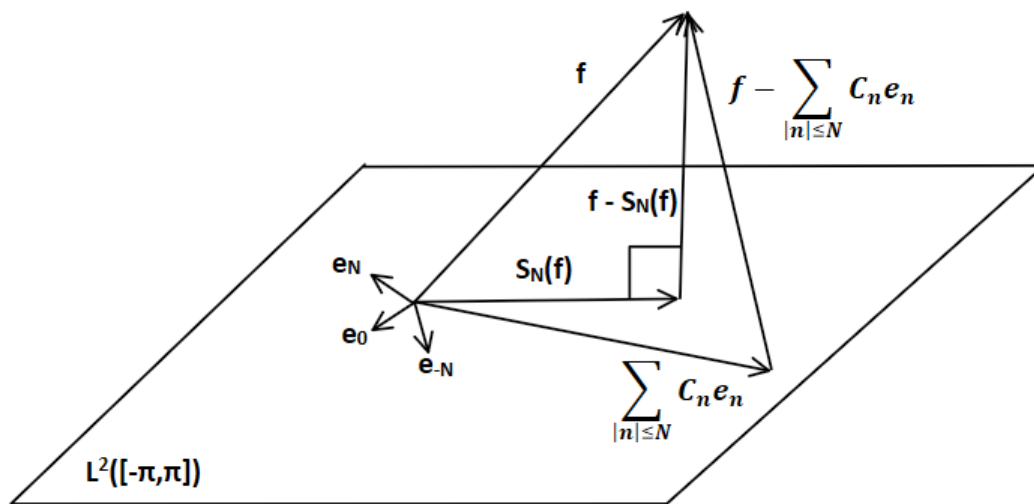
$$a_n = (f, e_n) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx.$$

In particular, this implies:

$$S_N(f) = \sum_{|n| \leq N} a_n e_n \implies \|S_N(f)\| = \left\| \sum_{|n| \leq N} a_n e_n \right\| = \sum_{|n| \leq N} |a_n|^2.$$

Then since the family $\{e_n\}$ is orthonormal, and since $a_n = (f, e_n)$, then indeed the difference $f - S_N(f)$ is orthogonal to e_n for all $|n| \leq N$, which implies our claim. $\square$

Now that we've proven our error function is orthogonal to the span of our basis, recall geometrically how the shortest path between a point and a plane is a vector orthogonal to the plane. In the same sense, we use our previous lemma to prove the previously described best approximation lemma. The intuition behind the best approximation lemma is easier to geometrically understand with the image below.



We now precisely state the functional version of the Best approximation lemma on $\mathbb{R}^n$ from linear algebra.

Lemma 4.2 (Best approximation lemma). *If f is integrable on the circle with Fourier coefficients a_n , then*

$$\|f - S_N(f)\| \leq \|f - \sum_{|n| \leq N} c_n e_n\|$$

for any complex numbers c_n . Moreover, equality holds precisely when $c_n = a_n$ for all $|n| \leq N$.

Proof :

Applying Pythagorean theorem:

$$f - S_N(f) + \sum_{|n| \leq N} b_n e_n = f - \sum_{|n| \leq N} c_n e_n, \quad \text{where } b_n = a_n - c_n.$$

Therefore as desired:

$$0 \leq \left\| \sum_{|n| \leq N} b_n e_n \right\| \implies \|f - \sum_{|n| \leq N} c_n e_n\| = \|f - S_N(f)\| + \left\| \sum_{|n| \leq N} b_n e_n \right\| \geq \|f - S_N(f)\|.$$

□

The orthogonality relationship from lemma 4.1 also implies an important inequality known as Bessel's inequality.

Theorem 4.3 (Bessel's inequality). *If f is integrable on the circle with $\{e_n\}$ being any orthonormal family of functions on the circle, and $a_n = (f, e_n)$, then:*

$$\sum_{n=-\infty}^{\infty} |a_n|^2 \leq \|f\|^2.$$

Further, when $\{e_n\}$ is also a basis and a_n are the Fourier coefficients of f , then equality holds. In the case of equality, the inequality becomes Parseval's identity, an important connection between the norms of the vector spaces $l^2(\mathbb{Z})$ and $L^2([-\pi, \pi])$.

Notice Parseval's identity is like an infinite-dimensional analogue to Pythagorean theorem in two dimensions. Here $\|f\|$ takes the place of the length of the hypotenuse c , and each $|a_n|$ takes the place of the length of the remaining two orthogonal legs a and b .

4.2. Mean-Square Convergence.

Toward proving the Mean-square convergence theorem, we use the following two theorems without proof.

Lemma 4.4. *Suppose f is integrable on the circle and bounded by B . Then there exists a sequence $\{f_k\}_{k=1}^{\infty}$ of continuous functions on the circle so that*

$$\sup_{x \in [-\pi, \pi]} |f_k(x)| \leq B \quad \text{for all } k = 1, 2, \dots,$$

and:

$$\int_{-\pi}^{\pi} |f(x) - f_k(x)| dx \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Theorem 4.5. *Continuous functions on the circle can be uniformly approximated by trigonometric polynomials.*

This means if f is continuous on $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$ and $\epsilon > 0$, then there exists a trigonometric polynomial P such that:

$$|f(x) - P(x)| < \epsilon \quad \forall x \in [-\pi, \pi].$$

This is the periodic analogue of the Weierstrass approximation theorem we state in section 5.3 for polynomials. Now we are ready to state the mean-square convergence theorem.

Theorem 4.6 (Mean-square convergence theorem). *Suppose f is integrable on the circle. Then:*

$$\frac{1}{2\pi} \int_0^{2\pi} |f(x) - S_N(f)(x)|^2 dx \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

In other words, f converges to S_N in L^2 -norm.

Proof : Suppose $f \in C([0, 2\pi])$. Given $\epsilon > 0$, by theorem 4.5 there exists a trigonometric polynomial P , say of degree M , such that:

$$|f(x) - P(x)| < \epsilon \quad \forall x \in [0, 2\pi].$$

Taking squares and integrating this inequality yields $\|f - P\| < c_1\epsilon$, for some constant c_1 . Then by lemma 4.2:

$$\|f - S_N\| \leq \|f - P\| < c_1\epsilon \quad \text{whenever } N \geq M.$$

This proves theorem 4.6 when f is continuous.

If f is merely integrable, apply lemma 4.4 to choose $g \in C([0, 2\pi])$ s.t. for some constant A_1

$$\sup_{x \in [0, 2\pi]} |g(x)| \leq \sup_{x \in [0, 2\pi]} |f(x)| = A_1,$$

and:

$$\int_0^{2\pi} |f(x) - g(x)| dx < \epsilon^2.$$

Then we may estimate that for some constant A_2 :

$$\begin{aligned} \|f - g\|^2 &= \frac{1}{2\pi} \int_0^{2\pi} |f(x) - g(x)|^2 dx \\ &\leq \frac{2A_1}{2\pi} \int_0^{2\pi} |f(x) - g(x)| dx \\ &\leq A_2 \epsilon^2. \end{aligned}$$

We may again use theorem 4.5 to approximate g by a trigonometric polynomial P s.t. $\|g - P\| < \epsilon$. Then $\|f - P\| < A_3\epsilon$ for some constant A_3 . Apply lemma 4.2 again and Pythagorean theorem to conclude the proof:

$$\|f - P\| \leq \|f - g\| + \|g - P\| < A_3\epsilon.$$

□

4.3. Pointwise Convergence.

We've shown that all we need for a function to converge to its Fourier Series in the L^2 -norm is integrability. We now examine if we can do better by achieving convergence in the Euclidean norm between a function evaluated at any point on the circle and the function's Fourier series evaluated at the same point, i.e. pointwise convergence. We will show that if f is also differentiable at x_0 , then its Fourier series evaluated at x_0 converges to $f(x_0)$.

We begin by establishing one lemma, one definition, and two propositions needed in the proof of the pointwise convergence theorem.

Lemma 4.7 (Riemann-Lebesgue lemma). *If f is integrable on the circle, then $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$.*

Equivalently, we may state the Riemann-Lebesgue lemma as follows; If f is integrable on the circle, then:

$$\int_0^{2\pi} f(x) \sin(Nx) dx \rightarrow 0 \quad \text{and} \quad \int_0^{2\pi} f(x) \cos(Nx) dx \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

The proof is out of the scope of this paper.

Definition 4.8. Define the N^{th} **Dirichlet kernel** as:

$$D_N(t) = \sum_{|n| \leq N} e^{inx}.$$

The following two important propositions follow immediately from the definition of the Dirichlet kernel.

Proposition 4.9. *If f is an integrable function on the circle, then $S_N(f)(x_0) = (f * D_N)(x_0)$.*

Proof :

$$\begin{aligned} S_N(f)(x) &= \sum_{|n| \leq N} \hat{f}(n) e^{inx} = \sum_{|n| \leq N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \right) e^{inx} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(\sum_{|n| \leq N} e^{in(x-y)} \right) dy = (f * D_N)(x). \end{aligned}$$

□

Proposition 4.10. *For the Dirichlet kernel D_N , we have the following two identities:*

$$(4.1) \quad tD_N(t) = \frac{t}{\sin(t/2)} \sin((N + 1/2)t),$$

$$(4.2) \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(x) dx = 1.$$

Proof : First to prove (4.1), we may write the Dirichlet kernel as:

$$D_N(x) = e^{-iNx} \sum_{n=0}^{2N} e^{inx}.$$

Since $\sum_{n=0}^K r^n = (r^{K+1} - 1)/(r - 1)$ for any $r \neq 1$, for $x \neq 0$ we have:

$$\begin{aligned} D_N(x) &= e^{-iNx} \frac{e^{i(2N+1)x} - 1}{e^{ix} - 1} = \frac{e^{i(N+1)x} - e^{-iNx}}{e^{ix} - 1} \\ &= \frac{e^{i(N+1)x} - e^{-iNx}}{e^{ix} - 1} \frac{e^{-ix/2}}{e^{-ix/2}} = \frac{e^{i(N+1/2)x} - e^{-i(N+1/2)x}}{e^{ix/2} - e^{-ix/2}} = \frac{\sin(N + 1/2)x}{\sin(x/2)}. \end{aligned}$$

To prove (4.2), we use the following identity:

$$D_N(x) = \sum_{|n| \leq N} e^{inx} = e^0 + \frac{1}{\pi} \sum_{n=1}^N \frac{e^{inx} - e^{-inx}}{2} = 1 + 2 \sum_{n=1}^N \cos(nx).$$

Directly integrate to obtain our desired result:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 + 2 \sum_{n=1}^N \cos(nx) dx = \left[\frac{x}{2\pi} + \sum_{n=1}^N \frac{\sin nx}{n\pi} \right]_{-\pi}^{\pi} = 1.$$

□

Identity (4.1) is useful because continuous functions satisfy the integrability condition of (4.7). The quotient $\frac{t}{\sin(t/2)}$ achieves this. Identity (4.2) is useful because we will want our main integrand to include a difference quotient so we may use that f is given to be differentiable. Achieve this using that:

$$f(x_0) \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(t) dt \right) = f(x_0).$$

Theorem 4.11 (Pointwise convergence theorem). *Let f be an integrable function on the circle which is differentiable at a point x_0 . Then $S_N(f)(x_0) \rightarrow f(x_0)$ as N tends to infinity.*

Proof : We must estimate $S_N(f)(x) - f(x)$, then show $S_N(f)(x) - f(x) \rightarrow 0$ as $N \rightarrow \infty$. We achieve this by applying the Riemann-Lebesgue lemma to $S_N(f)(x) - f(x)$ after we show $S_N(f)(x) - f(x)$ can be shown to satisfy the condition to be integrable on the circle. First we use the Dirichlet kernel and (4.2) to

find that:

$$\begin{aligned}
 S_N(f)(x) - f(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0 - t) D_N(t) dt - f(x_0) = (f * D_N)(x_0) - f(x_0) \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0 - t) D_N(t) dt - \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0) D_N(t) dt \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(x_0 - t) - f(x_0)) D_N(t) dt.
 \end{aligned}$$

Rewrite $f(x_0 - t) - f(x_0) = (t) \frac{f(x_0 - t) - f(x_0)}{t}$. Deal with the $t = 0$ case by defining:

$$F(t) = \begin{cases} \frac{f(x_0 - t) - f(x_0)}{t} & 0 < |t| < \pi \\ -f'(x_0) & t = 0. \end{cases}$$

F is bounded near 0 since f is differentiable there. The given function f being integrable implies for any small δ , F will be integrable on $[-\pi, -\delta] \cup [\delta, \pi]$. The only remaining concern is if F is integrable when $|t| < \delta$. This can be easily resolved since we prove in Math Analysis II that for any arbitrarily small δ , removing the open interval $(-\delta, \delta)$ from our integral's domain does not change the integral value (see proposition 1.4 in appendix 1 of [1]). Therefore by substituting in F and using (4.1):

$$S_N(f)(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) t D_N(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) \frac{t}{\sin(t/2)} \sin((N + 1/2)t) dt.$$

Decompose $\sin((N + 1/2)t)$ with following the angle addition identity:

$$\sin((N + 1/2)t) = \sin(Nt) \cos(t/2) + \cos(Nt) \sin(t/2).$$

Hence we get:

$$\begin{aligned}
 S_N(f)(x) - f(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) \frac{t}{\sin(t/2)} \left(\sin(Nt) \cos(t/2) + \cos(Nt) \sin(t/2) \right) dt \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) \frac{t}{\sin(t/2)} \left(\sin(Nt) \cos(t/2) \right) \\
 &\quad + \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) t \cos(Nt).
 \end{aligned}$$

Finally, since the product of continuous functions are continuous then $\frac{t}{\sin(t/2)} \left(\sin(Nt) \cos(t/2) \right)$ is continuous and bounded on $[-\pi, \pi]$. This implies its product with another integrable function $F(t)$ is also integrable, so the first integrand is integrable. Similarly, $t \cos(Nt)$ is integrable and bounded since it is continuous on $[-\pi, \pi]$. Hence our second integrand, $F(t) t \cos(Nt)$, is the product of integrable functions, and therefore is integrable. Finally, we've satisfied the conditions to apply lemma 4.7 to $S_N(f)(x) - f(x)$, so $S_N(f)(x) - f(x) \rightarrow 0$ as $N \rightarrow \infty$ as desired. $\square$

This concludes our focus on Fourier series. For the remainder of this paper, we focus on the Fourier transform.

5. Fourier Transform and Its Properties

5.1. Fourier Transform on $\mathbb{R}$.

For the remainder of this paper, the majority of our functions we will be working with will be on $\mathbb{R}$ or $\mathbb{R}^d$. Furthermore, these functions will be at least of moderate decrease, if not rapidly decreasing. Thus, it is worth exploring where and how they may be used.

Theorem 5.1. For $f, g \in \mathcal{M}(\mathbb{R})$:

(1) *Linearity*: If $f, g \in \mathcal{M}(\mathbb{R})$ and $a, b \in \mathbb{C}$, then:

$$\int_{\mathbb{R}} (af(x) + bg(x))dx = a \int_{\mathbb{R}} f(x)dx + b \int_{\mathbb{R}} g(x)dx.$$

(2) *Translation invariance*: $\forall h \in \mathbb{R}$, we have:

$$\int_{\mathbb{R}} f(x - h)dx = \int_{\mathbb{R}} f(x)dx.$$

(3) *Scaling under dilations*: If $\delta > 0$, then:

$$\delta \int_{\mathbb{R}} f(\delta x)dx = \int_{\mathbb{R}} f(x)dx.$$

(4) *Continuity*: If $f \in \mathcal{M}(\mathbb{R})$, then:

$$\int_{\mathbb{R}} |f(x - h) - f(x)|dx \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Proof :

(1) *Linearity*: Follows directly from linearity of integrals.

(2) *Translational invariance*: It suffices to see that:

$$\int_{-N}^N f(x - h)dx - \int_{-N}^N f(x)dx \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Since f is of moderate decrease, then for some $A \in \mathbb{R}$, $|f(x)| \leq A$, $\forall x \in \mathbb{R}$. Since $\int_{-N}^N f(x - h)dx = \int_{-N-h}^{N-h} f(x)dx$, then for some $\epsilon > 0$:

$$\begin{aligned} \left| \int_{-N}^N f(x - h)dx - \int_{-N}^N f(x)dx \right| &\leq \left| \int_{-N-h}^{-N} f(x)dx \right| + \left| \int_{N-h}^N f(x)dx \right| \\ &\leq \int_{-N}^N \frac{2A}{(1+x)^{1+\epsilon}} \leq \frac{A'}{(1+N)^\epsilon} \rightarrow 0, \quad \text{as } N \rightarrow \infty. \end{aligned}$$

(3) *Scaling under dilations*: We skip the proof since it is very similar to that of the previous property noticing that $\delta \int_{-N}^N f(\delta x)dx = \int_{-\delta N}^{\delta N} f(x)dx$.

(4) *Continuity*: It suffices to take $|h| \leq 1$. Given any fixed $\epsilon > 0$, choose N sufficiently large such that:

$$\int_{|x| \geq N} |f(x)|dx \leq \epsilon/4 \quad \text{and} \quad \int_{|x| \geq N} |f(x - h)|dx \leq \epsilon/4.$$

Since f is continuous on $[-N-1, N+1]$, then it is also uniformly continuous on the same interval. Therefore $\sup_{|x| \leq N} |f(x - h) - f(x)| \rightarrow 0$ as $h \rightarrow 0$, so we may take h to be sufficiently small s.t. this supremum is less than $\epsilon/4N$. Finally:

$$\int_{\mathbb{R}} |f(x - h) - f(x)|dx \leq \int_{-N}^N |f(x - h) - f(x)|dx + \int_{|x| \geq N} |f(x - h)|dx + \int_{|x| \geq N} |f(x)|dx \leq \epsilon/2 + \epsilon/4 + \epsilon/4.$$

□

Next, we prove an intuitive property of Schwartz functions, but one not inherent to their given definition. It will be used often to integrate Schwartz functions over $\mathbb{R}$ without diverging to infinity. Informally it states that for sufficiently large $|x|$, we can make the tail of the integral of any Schwartz function f arbitrarily small. Below it is written formally,

Proposition 5.2. *Given $f \in \mathcal{S}(\mathbb{R})$ and $\epsilon > 0$, $\exists N \in \mathbb{N}$ sufficiently large s.t. $\int_{|x| \geq N} |f(x)|dx < \epsilon$.*

Proof : By definition of a Schwartz function, for any given $f \in \mathcal{S}(\mathbb{R})$ and $k, l \geq 0$, $\exists A \in \mathbb{R}$ s.t. $\sup_{x \in \mathbb{R}} |x|^k |f^{(l)}(x)| \leq A < \infty$. Let $k > 1$, $l = 0$, and $x \neq 0$, $\exists A \geq 0$ s.t. $|f(x)| \leq \frac{A}{|x|^k}$. Integrate both sides of the previous inequality over the set $\{x : |x| \geq N\}$:

$$\int_{|x| \geq N} |f(x)| dx \leq \int_{|x| \geq N} \frac{A}{|x|^k} dx = 2A \frac{x^{1-k}}{1-k} \Big|_N^\infty = 2A \left| 0 - \frac{N^{1-k}}{1-k} \right| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

□

Now we are ready to define the Fourier transform. It can be thought of as the generalization of the idea of Fourier coefficients from discrete frequencies to a “continuous version”.

Definition 5.3. If $f \in \mathcal{M}(\mathbb{R})$, we define its **Fourier transform** for $\xi \in \mathbb{R}$ as:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

Notice the similarity between this and the formula for Fourier coefficients:

$$\hat{f}(n) = \int_{-L/2}^{L/2} f(x) e^{-2\pi i x n/L} dx.$$

We see ξ has replaced n/L . We will use the following properties of the Fourier transform in our major theorems:

Proposition 5.4. If $f \in \mathcal{S}(\mathbb{R})$ then,

- (1) *Translations become rotations:* $f(x+h) \rightarrow \hat{f}(\xi) e^{2\pi i h \xi}$ whenever $h \in \mathbb{R}$.
- (2) *Rotations become translations:* $f(x) e^{-2\pi i x h} \rightarrow \hat{f}(\xi + h)$ whenever $h \in \mathbb{R}$.
- (3) *Scaling under dilations:* $f(\delta x) \rightarrow \delta^{-1} \hat{f}(\delta^{-1} \xi)$ whenever $\delta > 0$.
- (4) *Differentiation becomes multiplication by $2\pi i \xi$:* $f'(x) \rightarrow 2\pi i \xi \hat{f}(\xi)$.
- (5) *Multiplication by $2\pi i x$ becomes differentiation:* $-2\pi i x f(x) \rightarrow \frac{d}{d\xi} \hat{f}(\xi)$.

Proof :

- (1) Translations become rotations: By translation invariance:

$$f(x+h) \rightarrow \int_{\mathbb{R}} f(x+h) e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} e^{2\pi i h \xi} dx = \hat{f}(\xi) e^{2\pi i h \xi}.$$

- (2) Rotations become translations:

$$f(x) e^{-2\pi i x h} \rightarrow \int_{\mathbb{R}} f(x+h) e^{-2\pi i x \xi} dx = \left(\int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx \right) e^{2\pi i h \xi} = \hat{f}(\xi) e^{2\pi i h \xi}.$$

- (3) Scaling under dilations: By the scaling under dilations property of functions of moderate decrease:

$$f(\delta x) \rightarrow \int_{\mathbb{R}} f(\delta x) e^{-2\pi i x \xi} dx = \delta^{-1} \int_{\mathbb{R}} f(x) e^{-2\pi i x \delta^{-1} \xi} dx = \delta^{-1} \hat{f}(\delta^{-1} \xi).$$

- (4) Differentiation becomes multiplication by $2\pi i \xi$: Integrate by parts, differentiating $e^{-2\pi i x \xi}$ and integrating $f'(x)$:

$$f'(x) \rightarrow \lim_{N \rightarrow \infty} \int_{-N}^N f'(x) e^{-2\pi i x \xi} dx = \lim_{N \rightarrow \infty} f(x) e^{-2\pi i x \xi} \Big|_{-N}^N + 2\pi i \xi \int_{-N}^N f(x) e^{-2\pi i x \xi} dx.$$

We know $\lim_{N \rightarrow \infty} f(x) e^{-2\pi i x \xi} \Big|_{-N}^N = 0$, since f decays rapidly and $|e^{-2\pi i x \xi}| = 1$.

Finally, we get that $\lim_{N \rightarrow \infty} 2\pi i \xi \int_{-N}^N f(x) e^{-2\pi i x \xi} dx = 2\pi i \xi \hat{f}(\xi)$.

- (5) Multiplication by $2\pi ix$ becomes differentiation: We use the difference quotient of f to show that $\hat{f}$ is differentiable and show its derivative is $-2\pi ix f(x)$. Let $\epsilon > 0$ be given. Consider:

$$\begin{aligned}
 \frac{\hat{f}(\xi + h) - \hat{f}(\xi)}{h} - (-2\pi ix f)(\xi) &= \int_{-\infty}^{\infty} f(x) e^{-2\pi ix \xi} \left[\frac{e^{-2\pi ix h} - 1}{h} + 2\pi ix \right] dx \\
 &\leq \int_{-\infty}^{\infty} |f(x) e^{-2\pi ix \xi} \left[\frac{e^{-2\pi ix h} - 1}{h} + 2\pi ix \right]| dx \\
 &\leq 2 \int_{-\infty}^{\infty} \left| \frac{f(x)}{h} \right| + \pi |x f(x)| dx \\
 (5.1) \quad &= 2 \int_{|x| < N} \left| \frac{f(x)}{h} \right| + \pi |x f(x)| dx \\
 (5.2) \quad &+ 2 \int_{|x| \geq N} \left| \frac{f(x)}{h} \right| + \pi |x f(x)| dx.
 \end{aligned}$$

We consider the head (5.1) and tail (5.2) separately. The first advantage to our decomposition is that f rapidly decreasing reduces finding the tail of the integral to a simple application of proposition 5.2. The second advantage is the head becomes an integral over a finite domain.

First considering the tail, by the same argument used in the proof of proposition 5.2 we know $\exists N$ sufficiently large such that:

$$\int_{|x| \geq N} |x| |f(x)| dx < \epsilon.$$

Therefore the previous fact along with proposition 5.2 implies the tail can be made arbitrarily small. Now considering the head, let $g(h) = e^{-2\pi ix h} \implies g'(0) = 2\pi ix$. Therefore, $\exists h_0 > 0$ s.t. $|h| < h_0$ implies:

$$\left| \frac{g(0) - g(h)}{0 - h} - g'(0) \right| = \left| \frac{e^{-2\pi ix h} - 1}{h} + 2\pi ix \right| < \frac{\epsilon}{2N}, \quad \forall N \in \mathbb{N}.$$

Hence for $|h| < h_0$ and for some positive constant C we have:

$$\int_{-N}^N \left| f(x) e^{-2\pi ix \xi} \left[\frac{e^{-2\pi ix h} - 1}{h} + 2\pi ix \right] \right| dx < C\epsilon.$$

Therefore both the head and tail can be made arbitrarily small with a sufficiently large N , so their sum is also arbitrarily small as desired. □

Propositions 5.4, (4) and 5.4, (5) tells us we may interchange multiplication (except factors of $2\pi i$) and differentiation. This is a key property we will use often, including the following two proofs below and the proof of Heisenberg's uncertainty principle. Another important property is that for any given Schwartz function, the Fourier transform preserves the rapid decrease property.

Theorem 5.5. *If $f \in \mathcal{S}(\mathbb{R})$, then $\hat{f} \in \mathcal{S}(\mathbb{R})$.*

Proof : By propositions 5.4, (4) and 5.4, (5), given $f \in \mathcal{S}(\mathbb{R})$, $\forall l, k \in \mathbb{N}$, $\xi^k \left(\frac{d}{d\xi} \right)^l \hat{f}(\xi)$ is the Fourier transform of $\frac{1}{(2\pi i)^k} \left(\frac{d}{dx} \right)^k [(-2\pi ix)^l f(x)]$. □

An immediate application of the previous theorem is that we can easily determine that $f(x) = e^{-|x|}$ is not Schwartz since $e^{-|x|} \rightarrow \frac{2}{1 + \xi^2} = \hat{f}(\xi)$, and the product of functions $\xi^3 \hat{f}(\xi)$ is an unbounded function and hence $\hat{f}$ cannot be Schwartz.

After defining the Fourier transform, it is natural to question if there is an inverse Fourier transform. The next section will answer in the affirmative with the Fourier inversion formula.

5.2. Fourier Inversion Formula.

We build up to the Fourier inversion formula as well as the multiplication formula, which is nearly as important, by defining a family of good kernels of the form $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$ and proving relevant theorems. It may be difficult to see why these theorems are useful until they are used in following proofs, and unfortunately there is no good intuition behind the use of these theorems. The reader may find it helpful to skim this section in reverse order, tracking from the Fourier inversion formula how the previous theorems are used.

We begin by showing in the next theorem that the special case of our family of good kernels that if $\delta = 1$, then $K_\delta = \widehat{K}_\delta$. This equality will not be true for any other choice of $\delta > 0$. This is particularly helpful in conjunction with our transformation identities from the previous section, since it gives us a good base equation that we may transform. To prove this theorem, we need the following proposition,

Proposition 5.6. $\int_{\mathbb{R}} e^{-\pi x^2} dx = 1.$

Proof : Let $f(x) = e^{-\pi x^2}$. Proceed by squaring the integral to obtain a two-dimensional integral, then convert to polar coordinates. Define

$$F(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x \xi} dx,$$

and calculate:

$$\begin{aligned} [F(0)]^2 &= \left(\int_{\mathbb{R}} e^{-\pi x^2} dx \right)^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-\pi r^2} r dr d\theta = \int_0^{\infty} 2\pi r e^{-\pi r^2} dr = -e^{-\pi r^2} \Big|_0^{\infty} = 1. \end{aligned}$$

□

Theorem 5.7. *If $f(x) = e^{-\pi x^2}$, then $\hat{f}(\xi) = f(\xi)$.*

Proof : By proposition 5.4, (5), and the fact that $f'(x) = -2\pi x f(x)$, we obtain:

$$F'(\xi) = \int_{-\infty}^{\infty} f(x)(-2\pi i x) e^{-2\pi i x \xi} dx = i \int_{-\infty}^{\infty} f'(x) e^{-2\pi i x \xi} dx.$$

By proposition 5.4, (4), $F'(\xi) = i(2\pi i \xi) \hat{f}(\xi) = -2\pi \xi F(\xi)$. If we define $G(\xi) = F(\xi) e^{\pi \xi^2}$, then it follows that $G'(\xi) = 0$, hence G is constant. Since $F(0) = 1$, we conclude that G is identically equal to 1, therefore $F(\xi) = e^{-\pi \xi^2}$ as desired. □

In general for any good kernel, (not just when K_δ follows the form given above,) the three properties below must be true.

Definition 5.8. A family of kernels $\{K_\delta(x)\}_{\delta=1}^{\infty}$ on $\mathbb{R}$ is said to be a family of **good kernels** if it satisfies the following properties:

(a) For every $\delta > 0$:

$$\int_{\mathbb{R}} K_\delta(x) dx = 1.$$

(b) There exists $M > 0$ such that $\forall \delta > 0$:

$$\int_{\mathbb{R}} |K_\delta(x)| dx \leq M.$$

(c) For every $\eta > 0$:

$$\int_{|x|>\eta} |K_\delta(x)| dx \rightarrow 0, \quad \text{as } \delta \rightarrow 0.$$

It immediately follows that we may find the general form of $\widehat{K}_\delta(\xi)$ as shown in the next corollary.

Corollary 5.9. *If $\delta > 0$ and $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\widehat{K}_\delta(\xi) = e^{-\pi\delta\xi^2}$.*

Proof : From the result of theorem 5.7, apply proposition 5.4, (3) with δ replaced by $\delta^{-1/2}$. $\square$

This family will serve as an example of the Heisenberg uncertainty principle in section 6.1 by seeing the duality between K_δ and its transform using several different delta values. Until now, we've claimed $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$ is a family of good kernels without justification. The claim is proven below.

Theorem 5.10. *The collection $\{K_\delta\}_{\delta>0}$ is a family of good kernels as $\delta \rightarrow 0$.*

Proof : To prove property (a) from definition 5.8 we may change variables to $u = \delta^{-1/2}x$:

$$\widehat{K}_\delta(\xi) = \int_{\mathbb{R}} \delta^{-1/2} e^{-\pi x^2/\delta} e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} e^{-\pi u^2} e^{-2\pi i u \delta^{1/2} \xi} du.$$

By proposition 5.6, $\widehat{K}_\delta(0) = \int_{\mathbb{R}} e^{-\pi u^2} du = 1$. Since $K_\delta \geq 0$, it is clear that (b) is also true. Finally to prove (c) we can again change variables to $y = \delta^{-1/2}x$:

$$\int_{|x|>\eta} |K_\delta(x)| dx = \int_{|x|>\eta} \delta^{-1/2} |e^{-\pi x^2/\delta}| dx = \int_{|y|>\eta/\delta^{1/2}} e^{-\pi y^2} dy \rightarrow 0 \text{ as } \delta \rightarrow 0.$$

$\square$

Corollary 5.11. *If $f \in \mathcal{S}(\mathbb{R})$, then $(f * K_\delta)(x) \rightarrow f(x)$ uniformly in x as $\delta \rightarrow 0$.*

Proof : First, we claim that f is uniformly continuous on $\mathbb{R}$. Indeed by proposition 5.2, given $\epsilon > 0$, $\exists R > 0$ s.t. $|f(x)| < \epsilon/4$ whenever $|x| \geq R$. Moreover, f is continuous, hence uniformly continuous on the compact interval $[-R, R]$, and together with the previous observation, we can find $\eta > 0$ s.t. $|f(x) - f(y)| \leq |f(x)| + |f(y)| < \epsilon/4 + \epsilon/4 < \epsilon$ whenever $|x - y| < \eta$. Now we argue as usual. Using the first property of good kernels, we can write:

$$(f * K_\delta)(x) - f(x) = (f * K_\delta)(x) - f(x) \int_{\mathbb{R}} K_\delta(t) dt = \int_{\mathbb{R}} K_\delta(t) [f(x - t) - f(x)] dt.$$

Since $K_\delta \geq 0$, we find:

$$|(f * K_\delta)(x) - f(x)| \leq \int_{|t|>\eta} K_\delta(t) |f(x - t) - f(x)| dt + \int_{|t|\leq\eta} K_\delta(t) |f(x - t) - f(x)| dt.$$

The first integral is small by the third property of good kernels, and the fact that f is bounded, while the second integral is also small since f is uniformly continuous and $\int K_\delta = 1$. $\square$

Proposition 5.12 (Multiplication formula). *If $f, g \in \mathcal{S}(\mathbb{R})$, then:*

$$\int_{\mathbb{R}} f(x) \hat{g}(x) dx = \int_{\mathbb{R}} \hat{f}(y) g(y) dy$$

Proof : Suppose $F(x, y)$ is a continuous function in the plane $(x, y) \in \mathbb{R}^2$. Assume $|F(x, y)| \leq A/(1 + x^2)(1 + y^2)$. Then, for each x the function $F(x, y)$ is of moderate decrease in y , and similarly for each fixed y the function $F(x, y)$ is of moderate decrease in x . Moreover, the function $F_1(x) = \int_{\mathbb{R}} F(x, y) dy$ is continuous and of moderate decrease; similarly for $F_2(y) = \int_{\mathbb{R}} F(x, y) dx$. By Fubini's theorem:

$$(5.3) \quad \int_{\mathbb{R}} F_1(x) dx = \int_{\mathbb{R}} F_2(y) dy.$$

Now apply (5.3) to $F(x, y) = f(x)g(y)e^{-2\pi i xy}$. Then $F_1(x) = f(x)\hat{g}(x)$, and $F_2(y) = \hat{f}(y)g(y)$ yields our desired result:

$$\int_{\mathbb{R}} f(x) \hat{g}(x) dx = \int_{\mathbb{R}} \hat{f}(y) g(y) dy.$$

$\square$

Theorem 5.13 (Inversion formula). *If $f \in \mathcal{S}(\mathbb{R})$:*

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

Proof : We prove first $f(0) = \int_{-\infty}^{\infty} \hat{f}(\xi) d\xi$. Then to generalize let $F(y) = f(y+x)$ so $F(0) = f(x)$. Let $G_\delta(x) = e^{-\pi\delta x^2}$, let $K_\delta(x) = \delta^{-1/2} e^{-\pi x^2/\delta}$. Add and subtract $-\frac{\pi\delta\xi^2}{\delta^2}$ to complete the square, and then change variables to $u = x\delta^{1/2} + i\xi\delta^{-1/2}$ with $\delta^{-1/2} du = dx$:

$$\begin{aligned} \widehat{G}_\delta(\xi) &= \int_{\mathbb{R}} e^{-\pi\delta x^2} e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} e^{-\pi\delta(x^2 + \frac{2i\xi x}{\delta} - \frac{\xi^2}{\delta^2}) - \frac{\pi\delta\xi^2}{\delta^2}} dx \\ &= (e^{-\pi\xi^2/\delta}) \int_{\mathbb{R}} e^{-\pi(x\delta^{1/2} + i\xi\delta^{-1/2})^2} dx = K_\delta(\xi) \int_{\mathbb{R}} e^{-\pi u^2} du = K_\delta(\xi). \end{aligned}$$

By the multiplication formula:

$$\int_{-\infty}^{\infty} f(x) K_\delta(x) dx = \int_{-\infty}^{\infty} \hat{f}(\xi) G_\delta(\xi) d\xi.$$

First on the left-hand side, applying corollary 5.11 and that K_δ is even, as $\delta \rightarrow 0$:

$$(f * K_\delta)(0) = \int_{\mathbb{R}} f(x) K_\delta(0-x) dx = \int_{\mathbb{R}} f(x) K_\delta(x) dx \rightarrow f(0).$$

Now on the right-hand side:

$$\lim_{\delta \rightarrow 0} \int_{\mathbb{R}} \hat{f}(\xi) G_\delta(\xi) d\xi = \int_{\mathbb{R}} \hat{f}(\xi) \left(\lim_{\delta \rightarrow 0} G_\delta(\xi) \right) d\xi = \int_{\mathbb{R}} \hat{f}(\xi) d\xi.$$

Therefore using the two substitutions above, the multiplication formula becomes the claim we sought to prove:

$$f(0) = \int_{\mathbb{R}} \hat{f}(\xi) d\xi.$$

In general:

$$f(x) = F(0) = \int_{\mathbb{R}} \hat{F}(\xi) d\xi = \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

□

Hence we have rigorously defined the Fourier transform and derived its inverse transform to use in our two main results in sections 6.1 and 6.3.

5.3. Plancherel formula.

Before beginning our first main result proving the Heisenberg uncertainty principle, we need two more important theorems. The first is known as Plancherel formula which shows that the L^2 -norm of Schwartz functions is preserved by the Fourier transform. To prove Plancherel formula, we will need the following additional properties of the Fourier transform.

Proposition 5.14. *If $f, g \in \mathcal{S}(\mathbb{R})$, then $f * g \in \mathcal{S}(\mathbb{R})$.*

Proof : We first show in two cases that:

$$(5.4) \quad \sup_{x \in \mathbb{R}} \frac{|x|^l}{(1+|y|)^l} |g(x-y)| \leq \sup_{(x-y) \in \mathbb{R}} |x-y|^l |g(x-y)| \leq 2^l A_l.$$

Our first case is if $|x| \leq 2|y|$, then:

$$\sup_{x \in \mathbb{R}} \frac{|x|^l}{(1+|y|)^l} |g(x-y)| \leq \sup_{x \in \mathbb{R}} \frac{|2y|^l}{(1+|y|)^l} |g(x-y)| = \sup_{x \in \mathbb{R}} 2^l \frac{|y|^l}{(1+|y|)^l} |g(x-y)| < \sup_{x \in \mathbb{R}} 2^l |g(x-y)| \leq 2^l A_l.$$

Thus, (5.4) holds in the first case. Our second case is if $|x| > 2|y|$, then:

$$\frac{|x|}{1+|y|} < |x| = 2(|x| - \frac{|x|}{2}) < 2(|x| - |y|) \leq 2|x-y|.$$

This implies:

$$\sup_{x \in \mathbb{R}} \frac{|x|^l}{(1+|y|)^l} |g(x-y)| \leq \sup_{(x-y) \in \mathbb{R}} 2^l |x-y|^l |g(x-y)|.$$

By the definition of a Schwartz function, given any $l \geq 0$, then $g \in \mathcal{S}(\mathbb{R}) \implies \exists A_l \in \mathbb{R}$ such that:

$$\sup_{(x-y) \in \mathbb{R}} 2^l |x-y|^l |g(x-y)| \leq 2^l A_l$$

So (5.4) holds in the second case as well. Next we show:

$$\begin{aligned} \sup_x |x^l (f * g)(x)| &= \sup_x \left| x^l \int_{\mathbb{R}} f(y) g(x-y) dy \right| \\ &\leq \sup_x |x|^l \int_{\mathbb{R}} |f(y)| |g(x-y)| dy \leq A_l \int_{\mathbb{R}} |f(y)| (1+|y|)^l dy. \end{aligned}$$

To obtain the last inequality, we multiply both sides of (5.4) by $|f(y)|$, then integrate both sides with respect to y . Since f is Schwartz, the last expression is finite. Hence $x^l (f * g)(x)$ is a bounded function $\forall l \geq 0$.

These estimates carry over to the derivatives of $f * g$, thereby proving that $f * g \in \mathcal{S}(\mathbb{R})$ because:

$$(5.5) \quad \left(\frac{d}{dx} \right)^k (f * g)(x) = (f * \left(\frac{d}{dx} \right)^k g)(x) \quad \text{for } k \in \mathbb{N}.$$

We prove (5.5) first for $k = 1$. Define:

$$G_n(x) = f(y) \frac{g(x-y+1/n) - g(x-y)}{1/n}.$$

Then $f, g \in \mathcal{S}(\mathbb{R}) \implies \exists M \in \mathbb{R}$ s.t. $G_n(x) \leq M$. Let $G(x) = \lim_{n \rightarrow \infty} G_n(x) = f(y)g'(x-y)$. Then $|G_n(x)| \leq G(x)$, with $G(x)$ known to be integrable. Therefore:

$$\begin{aligned} \frac{d}{dx} (f * g) &= \frac{d}{dx} \int_{\mathbb{R}} f(y) g(x-y) dy \\ &= \lim_{h \rightarrow 0} \int_{\mathbb{R}} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \\ &= \int_{\mathbb{R}} f(y) g'(x-y) dy = (f * \left(\frac{d}{dx} \right) g)(x). \end{aligned}$$

We may pass the limit inside the integral due to the dominated convergence theorem proven in our real analysis class. By iteration, (5.5) follows for all k . □

Proposition 5.15. *If $f, g \in \mathcal{S}(\mathbb{R})$, then $f * g = g * f$.*

Proof : For any fixed x :

$$\begin{aligned} (f * g)(x) &= \int_{-\infty}^{\infty} f(y) g(x-y) dy \\ (5.6) \quad &= - \int_{\infty}^{-\infty} f(-z) g(x+z) dz \\ &= \int_{-\infty}^{\infty} f(-z) g(x+z) dz = \int_{-\infty}^{\infty} f(x-(z+x)) g(z+x) dz \\ (5.7) \quad &= \int_{-\infty}^{\infty} f(x-u) g(u) du = (g * f)(x). \end{aligned}$$

Equality (5.6) follows from the first change of variables to $z = -y$. Equality (5.7) follows from the second change of variables to $u = z + x$. $\square$

Notice as a byproduct, we have also shown that for any $F \in \mathcal{S}(\mathbb{R})$, we have that $\int_{\mathbb{R}} F(x)dx = \int_{\mathbb{R}} F(-x)$.

Proposition 5.16. $\widehat{(f * g)}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Proof : First we want to prove $F(x, y) = f(y)g(x - y)e^{-2\pi i x \xi}$ is of moderate decrease in both x and y . Without loss of generality, prove the result for x in three cases by showing for all $\epsilon > 0$ and for all fixed $y_0 \in \mathbb{R}$, there exists $A_{y_0} \in \mathbb{R}$ such that:

$$(5.8) \quad |f(y_0)g(x - y_0)e^{-2\pi i x \xi}| \leq \frac{A_{y_0}}{1 + |x|^{1+\epsilon}} \quad \forall x \in \mathbb{R}.$$

It suffices to show $\exists A_{y_0} \in \mathbb{R}$ such that:

$$(5.9) \quad |f(y_0)g(x - y_0)e^{-2\pi i x \xi}|(1 + |x|^{1+\epsilon}) \leq |f(y_0)g(x - y_0)|(1 + |x|^{1+\epsilon}) \leq A_{y_0} < \infty.$$

Our first case is if $|x| > 2|y_0| = 0$, then by the definition of $f, g \in \mathcal{S}(\mathbb{R})$:

$$|f(0)g(x)e^{-2\pi i x \xi}|(1 + |x|^{1+\epsilon}) \leq |f(0)||g(x)|(1 + |x|^{1+\epsilon}) \leq A_{y_0}.$$

Our second case is if $|x| > 2|y_0| > 0$. To obtain our bound, use that $g \in \mathcal{S}(\mathbb{R}) \implies \exists M_g \in \mathbb{R}$ s.t. $|g(x - y_0)||x - y_0| \leq M_g$. Similarly, $f \in \mathcal{S}(\mathbb{R}) \implies \exists M_f \in \mathbb{R}$ s.t. $|f(y_0)| \leq M_f$. Hence:

$$(5.10) \quad \begin{aligned} |f(y_0)g(x - y_0)|(1 + |x|)^{1+\epsilon} &\leq M_f |g(x - y_0)| \left(\frac{|x - y_0|}{|x - y_0|} \right)^{1+\epsilon} (1 + |x|)^{1+\epsilon} \\ &\leq M_f M_g \frac{(1 + |x|)^{1+\epsilon}}{|x - y_0|^{1+\epsilon}} = M_f M_g \frac{(1 + |x - y_0 + y_0|)^{1+\epsilon}}{|x - y_0|^{1+\epsilon}} \\ &\leq M_f M_g \left(\frac{|x - y_0| + |y_0|}{|x - y_0|} \right)^{1+\epsilon} \leq M_f M_g \left| 1 + \frac{y_0}{x - y_0} \right|^{1+\epsilon} \leq M_f M_g 2^{1+\epsilon} = A_{y_0}. \end{aligned}$$

We know (5.10) because:

$$|x - y_0| > |x| - |y_0| > 2|y_0| - |y_0| = |y_0| \implies \frac{1}{|x - y_0|} < \frac{1}{|y_0|} \implies \left| 1 + \frac{y_0}{x - y_0} \right|^{1+\epsilon} < 2^{1+\epsilon}.$$

Our third and final case is if $|x| \leq 2|y_0|$, then:

$$|f(y_0)||e^{-2\pi i x \xi}||g(x - y_0)|(1 + |x|)^{1+\epsilon} \leq M_f M_g (1 + |2y_0|)^{1+\epsilon} = A_{y_0}.$$

Hence with all three cases proven, we indeed know $F(x, y)$ is of moderate decrease in both x and y , so we may apply Fubini's theorem. The function $F_1(x) = \int_{\mathbb{R}} F(x, y)dy = (f * g)(x)e^{-2\pi i x \xi}$ is continuous and of moderate decrease, and similarly for $F_2(y) = \int_{\mathbb{R}} F(x, y)dx = f(y)e^{-2\pi i y \xi}\hat{g}(\xi)$. Thus we obtain our desired result:

$$\widehat{(f * g)}(\xi) = \int_{\mathbb{R}} (f * g)(x)e^{-2\pi i x \xi}dx = \int_{\mathbb{R}} F_1(x)dx = \int_{\mathbb{R}} F_2(y)dy = \left(\int_{\mathbb{R}} f(y)e^{-2\pi i y \xi}dy \right) \hat{g}(\xi) = \hat{f}(\xi)\hat{g}(\xi).$$

$\square$

Now we are ready to prove our first major theorem of this section.

Theorem 5.17 (Plancherel Formula). *If $f \in \mathcal{S}(\mathbb{R})$ then $\|\hat{f}\| = \|f\|$. (L^2 norm of Schwartz functions is preserved by the Fourier transform).*

Proof : Let $f^b(x) = \overline{f(-x)}$. Then since for any $F \in \mathcal{S}(\mathbb{R})$, $\int_{\mathbb{R}} F(x)dx = \int_{\mathbb{R}} F(-x)$, we may calculate the Fourier transform of f^b to be:

$$(5.11) \quad \widehat{f^b}(\xi) = \int_{\mathbb{R}} \overline{f(-x)}e^{-2\pi i x \xi}dx = \int_{\mathbb{R}} \overline{f(-x)e^{2\pi i x \xi}}dx = \int_{\mathbb{R}} \overline{f(x)e^{-2\pi i x \xi}}dx = \int_{\mathbb{R}} \overline{f(x)}e^{-2\pi i x \xi}dx = \overline{\hat{f}(\xi)}.$$

Let $h(x) = (f * f^b)(x)$. Then by proposition 5.16 and (5.11):

$$(5.12) \quad \hat{h}(\xi) = \hat{f}(\xi)\widehat{f^b}(\xi) = \hat{f}(\xi)\overline{\hat{f}(\xi)} = |\hat{f}(\xi)|^2.$$

Furthermore:

$$(5.13) \quad h(0) = (f * f^\flat)(0) = \int_{\mathbb{R}} f(t) f^\flat(0-t) dt = \int_{\mathbb{R}} f(t) f^\flat(0-t) dt = \int_{\mathbb{R}} f(t) \overline{f(t)} dt = \int_{-\infty}^{\infty} |f(t)|^2 dt.$$

By the Fourier inversion formula we get:

$$(5.14) \quad \int_{-\infty}^{\infty} \hat{h}(\xi) d\xi = \int_{-\infty}^{\infty} \hat{h}(\xi) e^{2\pi i \xi 0} d\xi = h(0).$$

Finally (5.12), (5.13), and (5.14) imply our result:

$$\|f\|^2 = \int_{\mathbb{R}} |\hat{f}(\xi)|^2 d\xi = \int_{\mathbb{R}} \hat{h}(\xi) d\xi = h(0) = \int_{\mathbb{R}} |f(t)|^2 dt = \|f\|^2.$$

□

We state the Weierstrass approximation theorem. We skip the proof (see chapter 5 of [1]).

Theorem 5.18 (Weierstrass approximation theorem). *Let f be a continuous function on the closed and bounded interval $[a, b] \subset \mathbb{R}$. Then $\forall \epsilon > 0$, $\exists$ polynomial P such that:*

$$\sup_{x \in [a, b]} |f(x) - P(x)| < \epsilon.$$

In other words, f can be uniformly approximated by polynomials.

Finally, to end the chapter we introduce new notation that will sparsely be used in the remainder of the paper. Denote the Fourier transform operator as $\mathcal{F}$, and similarly the inverse Fourier transform operator as $\mathcal{F}^{-1}$. Then $\mathcal{F}[f(x)] = \hat{f}(\xi)$ and $\mathcal{F}^{-1}[\hat{f}(\xi)] = f(x)$.

Now we have covered all necessary background knowledge to prove our first major application, the Heisenberg Uncertainty Principle.

5.4. Fourier Transform on $\mathbb{R}^d$.

The Fourier transform in $\mathbb{R}^d$ is not very different from the Fourier transform in $\mathbb{R}$. Yet, since we need the Radon transform in $\mathbb{R}^3$, then we must understand those differences to prove our main results in section 11. The central difference we explore in this section is that we may perform rotations in $\mathbb{R}^d$. In one-dimension, rotations are degenerate. Without loss of generality, assuming the axis of rotation is the origin, then rotation is simply a change of variables from x to $-x$. Yet we already been using their properties any time even or odd functions have been involved. Now we rigorously define rotations and the Fourier transform in $\mathbb{R}^d$.

Recall from linear algebra that rotations can be expressed as orthogonal matrices. We give the precise definition of a rotation below, but the general concept is the same as orthogonal matrices.

Definition 5.19. A **rotation** in $\mathbb{R}^d$ is a linear transformation $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ which preserves the inner product. In other words,

- (a). $R(ax + by) = aR(x) + bR(y)$ for all $x, y \in \mathbb{R}^d$ and $a, b \in \mathbb{R}$.
- (b). $R(x) \cdot R(y) = x \cdot y$ for all $x, y \in \mathbb{R}^d$.

Condition (b) that rotations preserve the inner product implies that for any rotation, R we must have $|\det(R)| = 1$. This should intuitively make sense since if we rotate a square in $\mathbb{R}^2$, we expect the area of the square to not change. The central property of functions of moderate decrease which immediately follows, and is often used to prove results concerning the Radon transform, is that for every rotation:

$$\int_{\mathbb{R}^d} f(R(x)) dx = \int_{\mathbb{R}^d} f(x) dx.$$

Now we've defined rotations, we are nearly ready to define the Fourier transform in $\mathbb{R}^d$. The functions relevant to the Radon transform are still of rapid decrease. It is apparent from definition 5.20 that $\mathcal{S}(\mathbb{R}^d)$ is nearly identical to $\mathcal{S}(\mathbb{R})$.

Definition 5.20. A continuous complex-valued function f on $\mathbb{R}^d$ is said to be **rapidly decreasing**, i.e. $f \in \mathcal{S}(\mathbb{R}^d)$, if for every multi-index α , the function $|x^\alpha f(x)|$ is bounded. Equivalently, a continuous function is of rapid decrease if:

$$\sup_{x \in \mathbb{R}^d} |x|^k |f(x)| < \infty \quad \forall k = 0, 1, 2, \dots$$

Definition 5.21. The **Fourier transform** of $f \in \mathcal{S}(\mathbb{R}^d)$ is defined by:

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx, \quad \text{for } \xi \in \mathbb{R}^d.$$

Each of the first five properties of the Fourier transform that we saw in $\mathbb{R}$ has a similar analogous property in $\mathbb{R}^d$. The final property concerns rotations and hence is only non-degenerate in $\mathbb{R}^d$.

Proposition 5.22. Let $f \in \mathcal{S}(\mathbb{R}^d)$,

- (1) $f(x + h) \rightarrow \hat{f}(\xi) e^{2\pi i \xi \cdot h}$ whenever $h \in \mathbb{R}^d$.
- (2) $f(x) e^{-2\pi i x h} \rightarrow \hat{f}(\xi + h)$ whenever $h \in \mathbb{R}^d$.
- (3) $f(\delta x) \rightarrow \delta^{-d} \hat{f}(\delta^{-1} \xi)$ whenever $\delta > 0$.
- (4) $\left(\frac{\partial}{\partial x}\right)^\alpha f(x) \rightarrow (2\pi i \xi)^\alpha \hat{f}(\xi)$.
- (5) $(-2\pi i x)^\alpha f(x) \rightarrow \left(\frac{\partial}{\partial \xi}\right)^\alpha \hat{f}(\xi)$.
- (6) $f(Rx) \rightarrow \hat{f}(R\xi)$ whenever R is a rotation.

Proof : Each of the first five proofs is the same as in the one-dimensional case. To prove the sixth and final property, change variables to $y = Rx$ in the integral. Since R is a rotation, then $|\det(R)| = 1$ and since R preserves the inner product, then $y \cdot \xi = Ry \cdot R\xi \implies R^{-1}y \cdot \xi = y \cdot R\xi$. Thus:

$$f(Rx) = f(y) \rightarrow \int_{\mathbb{R}^d} f(y) e^{-2\pi i (R^{-1}y) \cdot \xi} dy = \int_{\mathbb{R}^d} f(y) e^{-2\pi i y \cdot (R\xi)} dy = \hat{f}(R\xi).$$

□

Define a function f to be **radial** if it depends only on $|x|$. For example, take the Gaussian $e^{-\pi|x|^2}$. Observe that when $d = 1$, the definition of a radial function is equivalent to the definition of an even function. A helpful feature of radial functions follows.

Theorem 5.23. The Fourier transform of a radial function is radial.

Proof : If f is radial then $f(Rx) = f(x)$ for all rotations R and so by proposition 5.22, (6) similarly $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R , thus $\hat{f}$ is indeed radial whenever f is. □

In section 6, we will need the Fourier inversion formula in $\mathbb{R}^3$ as stated below. The proof is nearly identical to the one-dimensional version.

Theorem 5.24. If $f \in \mathcal{S}(\mathbb{R})$:

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi.$$

Now we are ready to prove our main results.

6. Important Applications: Heisenberg Uncertainty Principle and Radon Transform

6.1. Heisenberg Uncertainty Principle.

The Heisenberg uncertainty principle was discovered 1927 by German physicist Werner Heisenberg. It is often used in quantum mechanics. In the context of quantum mechanics, the uncertainty principle states

that given a particle, there is a fundamental positive lower bound to the precision of the measurement of position times the precision of the measurement of momentum of a particle. Hence, the more precisely we know one, the less precisely we know the other and vice-versa. It is commonly confused with the observer effect, which states that by observing the particle, its position and/or momentum is changed. The uncertainty principle applies to much more than simply physics. More generally, we define **complementary variables** as pairs of variables that share the same relationship described above as position and momentum. Now we precisely define the uncertainty principle.

Theorem 6.1. *Suppose $\psi \in \mathcal{S}(\mathbb{R})$ satisfies the normalizing condition $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$. Then*

$$(6.1) \quad \left(\int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} \xi^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2}$$

and equality holds if and only if $\psi(x) = Ae^{-Bx^2}$ where $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$.

In fact we have

$$(6.2) \quad \left(\int_{-\infty}^{\infty} (x - x_0)^2 |\psi(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} (\xi - \xi_0)^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2}$$

for every $x_0, \xi_0 \in \mathbb{R}$.

Let us define piece by piece the notation and physical interpretation behind the equations above. Without loss of generality, let us assume the uncertainty principle is being applied in the context of quantum mechanics. Notice we use some standard formulas from probability. $\psi(x)$ is a wave function, a solution of Schrodinger's equation. $|\psi(x)|^2$ is probability density function. $\int_a^b |\psi(x)|^2 dx$ gives the probability that $x \in (a, b)$, hence the normalizing condition $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$. If the expected position is measured as $x_0 = \mu_x = E[x]$, then $Var(x) = E[(x - x_0)^2] = \int_{\mathbb{R}} (x - x_0)^2 |\psi(x)|^2 dx$ is the mean-square error of our measurement of the position of a particle. Similarly, $\int_{\mathbb{R}} (\xi - \xi_0)^2 |\hat{\psi}(\xi)|^2 d\xi$ is the “mean-square error” of our measurement of the momentum of a particle.

Now we prove Heisenberg's uncertainty principle.

Proof :

We prove the theorem in three parts. We begin by proving (6.1). Assume the normalizing condition:

$$1 = \int_{\mathbb{R}} |\psi(x)|^2 dx.$$

Next, apply integration by parts, integrating dx and differentiating $|\psi(x)|^2$:

$$1 = x|\psi(x)|^2 \Big|_{-\infty}^{\infty} - \int_{\mathbb{R}} x \frac{d}{dx} |\psi(x)|^2 dx = - \int_{\mathbb{R}} x \frac{d}{dx} |\psi(x)|^2 dx.$$

Apply product rule to differentiate $|\psi|^2 = \psi\bar{\psi}$ and take absolute value:

$$1 = - \int_{\mathbb{R}} \left(x\psi'(x)\bar{\psi}(x) + x\bar{\psi}'(x)\psi(x) \right) dx \leq \int_{\mathbb{R}} |x\psi'(x)\bar{\psi}(x) + x\bar{\psi}'(x)\psi(x)| dx.$$

By Triangle inequality:

$$1 \leq \int_{\mathbb{R}} |x\psi'(x)\bar{\psi}(x) + x\bar{\psi}'(x)\psi(x)| dx \leq 2 \int_{\mathbb{R}} |x||\psi(x)||\bar{\psi}'(x)| dx.$$

Then by Cauchy-Schwartz inequality:

$$1 \leq 4 \int_{\mathbb{R}} \left(|x||\psi(x)||\bar{\psi}'(x)| \right)^2 dx \leq 4 \left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} |\psi'(x)|^2 dx \right).$$

By Plancherel formula:

$$1 \leq 4 \left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} |\psi'(x)|^2 dx \right) = 4 \left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} |\hat{\psi}'(\xi)|^2 d\xi \right).$$

Finally, Proposition 5.4, (4) implies (6.1):

$$1 \leq 4 \left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} |\hat{\psi}'(\xi)|^2 d\xi \right) = 16\pi^2 \left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} \xi^2 |\hat{\psi}(\xi)|^2 d\xi \right).$$

Equality holds in the Heisenberg uncertainty principle if and only if we have equality in the Cauchy-Schwartz inequality which implies linear dependence between our two functions i.e. $\psi'(x) = \beta x\psi(x)$ for some $\beta \in \mathbb{R}$. Notice this is an ODE. So we may solve this linear ODE using an integrating factor:

$$\psi(x) = Ae^{\beta x^2/2} \quad \text{for some } A \in \mathbb{C}.$$

Since $\psi \in \mathcal{S}(\mathbb{R})$ we know $\beta < 0$. Thus let $\beta = -2B$ for some constant $B > 0$. The normalizing condition implies:

$$1 = |A|^2 \int_{\mathbb{R}} e^{-2Bx^2} dx = |A|^2 \sqrt{\pi/2B} \implies |A|^2 = \sqrt{2B/\pi}.$$

Finally, we prove (6.2) as a consequence of (6.1). Given $\psi_1, \psi_2 \in \mathcal{S}(\mathbb{R})$ such that both satisfy the normalizing condition and $\psi_1(y) = \psi_2(y + x_0)e^{-2\pi iy\xi_0}$ for some constants $x_0, \xi_0 \in \mathbb{R}$, then by (6.1):

$$(6.3) \quad \left(\int_{-\infty}^{\infty} y^2 |\psi_1(y)|^2 dy \right) \left(\int_{-\infty}^{\infty} \gamma^2 |\hat{\psi}_1(\gamma)|^2 d\gamma \right) \geq \frac{1}{16\pi^2}.$$

To find $\hat{\psi}_1$ in terms of ψ_2 , we use the multiplication formula. Let $f(y) = \psi_2(y + x_0)$ and let $g(y) = e^{-2\pi iy\xi_0}$. Then $\widehat{f(y)g(y)} = \mathcal{F}[\psi_2(y + x_0)e^{-2\pi iy\xi_0}] = (\hat{f} * \hat{g})(\gamma)$. The identity from functional analysis $\mathcal{F}[g(y)] = \mathcal{F}[e^{-2\pi i\xi_0 y}] = \delta(\gamma + \xi_0)$ implies $\hat{\psi}_1(\gamma) = \hat{\psi}_2(\gamma + \xi_0)e^{2\pi ix_0(\gamma + \xi_0)}$. So substituting in $\psi_2(y)$, we get that (6.3) is equivalent to:

$$\left(\int_{-\infty}^{\infty} y^2 |\psi_2(y + x_0)e^{-2\pi iy\xi_0}|^2 dy \right) \left(\int_{-\infty}^{\infty} \gamma^2 |\hat{\psi}_2(\gamma + \xi_0)e^{2\pi ix_0(\gamma + \xi_0)}|^2 d\gamma \right) \geq \frac{1}{16\pi^2}.$$

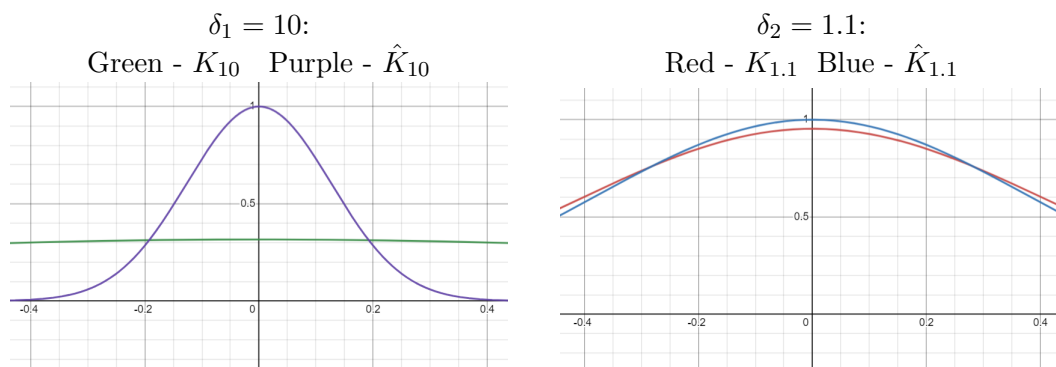
Finally use that $|e^{-2\pi i(x-x_0)\xi_0}| = 1$, and change variables letting $x = y + x_0$ and $\xi = \gamma + \xi_0$ to obtain (6.2):

$$\left(\int_{-\infty}^{\infty} (x - x_0)^2 |\psi_2(x - x_0)|^2 dx \right) \left(\int_{-\infty}^{\infty} (\xi - \xi_0)^2 |\hat{\psi}_2(\xi - \xi_0)|^2 d\xi \right) \geq \frac{1}{16\pi^2}.$$

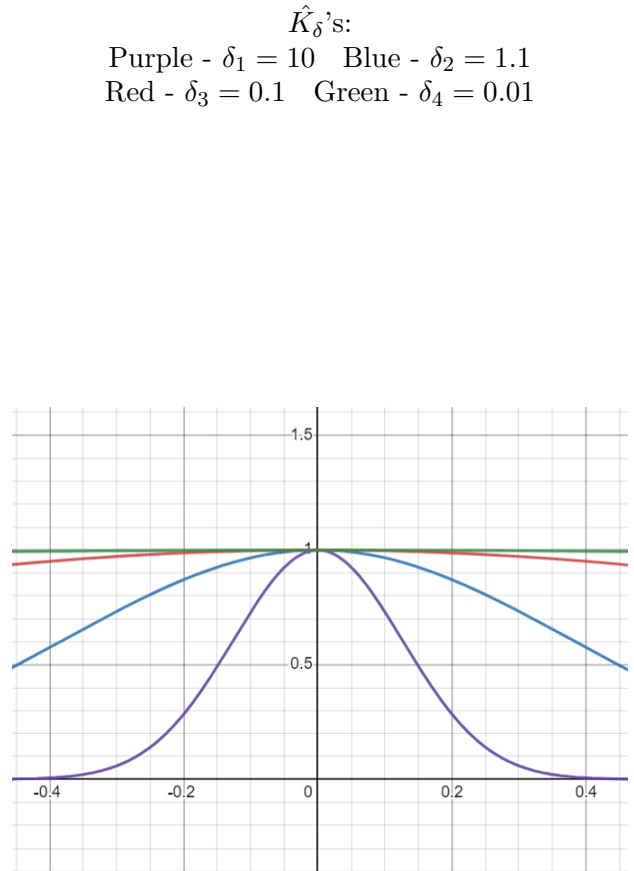
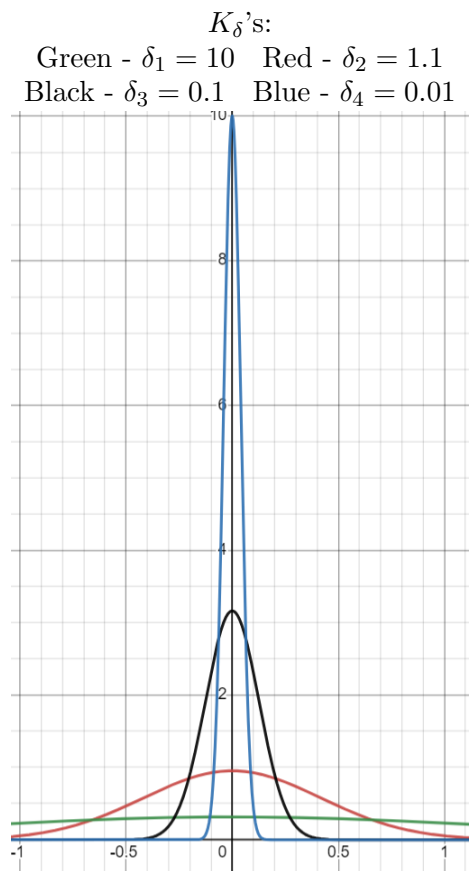
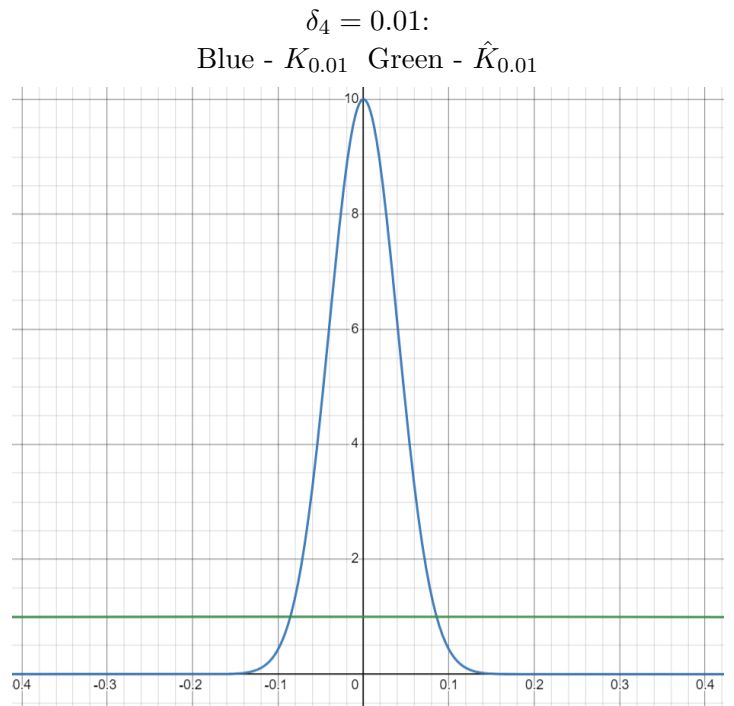
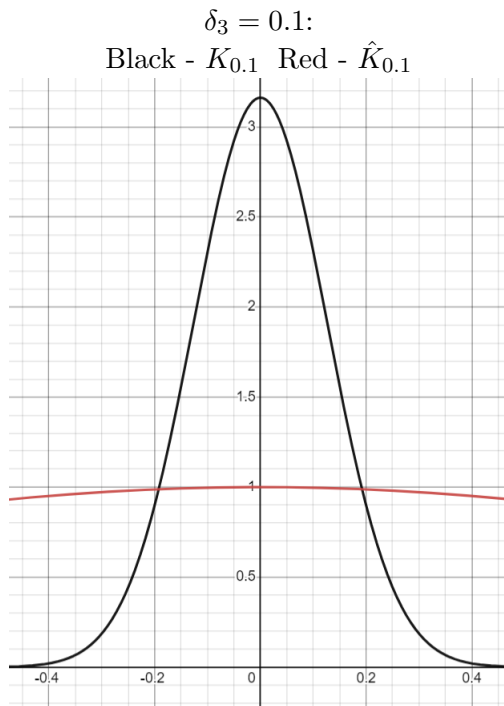
□

We may see an example¹ of the principle applied graphically using our family of good kernels from section 5.2. The images below demonstrate that as the graphs corresponding to K_δ become more concentrated at the origin, the graphs corresponding to $\hat{K}_\delta$ becomes less concentrated at the origin. Note that if $\delta = 1$, then $K_1 = \hat{K}_1$. We choose $\delta_2 = 1.1$ so the graphs don't overlap.

In our example, $K_\delta(x)$ will take the role of $|\psi(x)|^2$. We satisfy the normalizing condition since $K_\delta(x)$ is a good kernel ($\int_{\mathbb{R}} K_\delta(x) dx = 1$). Obtain the complementary variable of $K_\delta(x)$ with the Fourier transform. If $\delta > 0$, $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\hat{K}_\delta(\xi) = e^{-\pi\delta\xi^2}$. Notice As $\delta \rightarrow 0$, K_δ peaks at the origin, but as $\delta \rightarrow 0$, $\hat{K}_\delta$ becomes more flat. Therefore, clearly both functions cannot be concentrated at the origin.



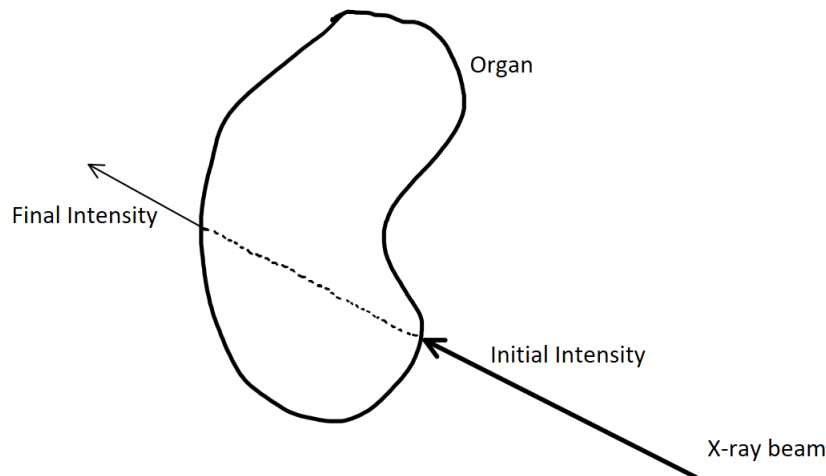
¹Another simple example of the principal applied could be the following physics problem: “Given we measure the expected momentum of an electron with a known uncertainty, and we measure the expected position, find the lower bound of uncertainty in our measurement of expected position of an electron”. To find the solution, simply divide our right hand side lower bound by the given uncertainty.



Now that we've proven our first main result, we build up theory on the Radon transform. This begins with expanding our definition of the Fourier from a single dimension to multiple in the next section.

6.2. Radon Transform.

The Radon transform was pioneered by Johann Radon in 1917. In the special case of $\mathbb{R}^2$, the Radon transform is often referred to as the X-ray transform. We however concern ourselves with the Radon transform in $\mathbb{R}^3$, the most applicable domain since humans experience three spacial dimensions. To gain intuition, imagine an organ being scanned with an X-ray.



Given the initial intensity of the X-ray beam, we measure the beam's intensity after it has passed through the organ. Now rotate your scanner and repeat until we've scanned the entire cross-section of an organ. The collected data from our scanner is a function, $X(\rho)$, the X-ray transform of ρ , the absorption coefficient function, for the cross-section of the organ. However, we desire data of the organ as a function ρ , not data of the intensity of the X-ray beam. The absorption coefficient function gives us useful information that for example, a medical doctor could use to perhaps evaluate if the given organ has cancer.

This leads to the natural questions of whether our transform X has an inverse transform and if the transform is unique. To answer the latter, we prove uniqueness with theorem 6.6. It turns out that the answer to the former is no, there is no inverse X-ray transform. However, we prove with theorem 6.8 that we may reconstruct ρ using the closely related dual Radon transform we define in the next section, and using the Laplacian operator.

In practice, this is all done by computer programs numerically using the Fast Fourier transform. The solution to the reconstruction problem contributed to rapid development of computerized imaging by sections using penetrative waves known as tomography.

Now that we have motivated our study of the Radon transform, we set to precisely defining it. The Radon Transform is an integral across a plane. Hence, we will use the following notation for planes in $\mathbb{R}^3$.

Definition 6.2. Let S^2 denote the unit sphere $\{x \in \mathbb{R}^3 : |x| = 1\}$. Given a unit vector $\gamma \in S^2$ and a number $t \in \mathbb{R}$, we define the **plane** $\mathcal{P}_{t,\gamma}$ by:

$$\mathcal{P}_{t,\gamma} = \{x \in \mathbb{R}^3 : x \cdot \gamma = t\}$$

Now we may make sense of a plane integral more precisely. We will need unit vectors $\{\gamma, e_1, e_2\}$ chosen to be an orthonormal basis for $\mathbb{R}^3$. Then for any $x \in \mathcal{P}_{t,\gamma}$ can be written uniquely as $x = t\gamma + u$, with $u = u_1e_1 + u_2e_2$ and $u_1, u_2 \in \mathbb{R}$. Finally, we can define the Radon transform.

Definition 6.3. The **Radon transform** of a function $f \in \mathcal{S}(\mathbb{R}^3)$ is defined by:

$$\mathcal{R}(f)(t, \gamma) = \int_{\mathcal{P}_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1e_1 + u_2e_2) du_1 du_2.$$

This transform has the domain and range of $\mathcal{R} : \mathcal{S}(\mathbb{R}^3) \rightarrow \mathcal{S}(\mathbb{R} \times S^2)$.

One necessary component of the Radon transform's uniqueness is for the transform's value to be independent of the orthogonal basis being used. The theorem below achieves this.

Theorem 6.4. *If $f \in \mathcal{S}(\mathbb{R}^3)$, then for each γ the definition of $\int_{\mathcal{P}_{t,\gamma}} f$ is independent of the choice of e_1 and e_2 . Moreover:*

$$\int_{-\infty}^{\infty} \left(\int_{\mathcal{P}_{t,\gamma}} f \right) dt = \int_{\mathbb{R}^3} f(x) dx.$$

Proof : To prove that our definition 6.3 is independent of the choice of e_1 and e_2 , we exploit that our Jacobian, $|\det(R)| = 1$, hence changing variables $u' = R(u)$ in the integral will not change the value of the integral. Therefore, if e'_1, e'_2 is another choice of basis vectors so that $\{\gamma, e'_1, e'_2\}$ is an orthonormal basis of $\mathbb{R}^3$, consider the rotation R in $\mathbb{R}^2$ which takes e_1 to e'_1 and e_2 to e'_2 . Since the value of the integral does not change, this proves that indeed our definition 6.3 is independent of the choice of basis as desired for part 1.

To prove the formula, let R denote the rotation which takes the standard basis of unit vectors in $\mathbb{R}^3$ to γ, e_1, e_2 . Then:

$$\int_{\mathbb{R}^3} f(x) dx = \int_{\mathbb{R}^3} f(Rx) dx = \int_{\mathbb{R}^3} f(x_1\gamma + x_2e_1 + x_3e_2) dx_1 dx_2 dx_3 = \int_{\mathbb{R}} \left(\int_{\mathcal{P}_{t,\gamma}} f \right) dt.$$

□

One may notice that so far in this section, the Fourier transform has not once been mentioned. In the following lemma, we see the important connection of the Radon transform with the Fourier transform.

Lemma 6.5. *If $f \in \mathcal{S}(\mathbb{R}^3)$ then $\mathcal{R}(f)(t, \gamma) \in \mathcal{S}(\mathbb{R})$ for each fixed γ . Moreover:*

$$\widehat{\mathcal{R}(f)}(s, \gamma) = \hat{f}(s\gamma).$$

Proof : Let γ be given and remain fixed throughout. Since f is Schwartz, then $\forall N \in \mathbb{N}, \exists A_N < \infty$ such that:

$$(6.4) \quad (1 + |t|)^N (1 + |u|)^N |f(t\gamma + u)| \leq A_N \implies (1 + |t|)^N |f(t\gamma + u)| \leq \frac{A_N}{(1 + |u|)^N}.$$

Integrate both sides with respect to u :

$$(1 + |t|)^N \mathcal{R}(f)(t, \gamma) \leq \int_{\mathbb{R}^2} \frac{A_N}{(1 + |u|)^N} du.$$

Our right hand side integral above is finite only when $N \geq 3$:

$$\int_{\mathbb{R}^2} \frac{A_N}{(1 + |u|)^N} du < \infty \quad \text{for } N \geq 3.$$

Hence if $N \geq 3$, then $\lim_{|u| \rightarrow \infty} \frac{A_N}{(1 + |u|)^N} = 0$, then indeed the product of any polynomial of u and t with $\mathcal{R}(f)$ is still bounded. Therefore $\mathcal{R}(f)(t, \gamma) \in \mathcal{S}(\mathbb{R})$.

To show $\widehat{\mathcal{R}(f)}(s, \gamma) = \hat{f}(s\gamma)$, by the definition of the Radon transform:

$$\begin{aligned} \widehat{\mathcal{R}(f)}(s, \gamma) &= \int_{\mathbb{R}} \left(\int_{\mathcal{P}_{t,\gamma}} f \right) e^{-2\pi i s t} dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u) du e^{-2\pi i s t} dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u) du e^{-2\pi i s \gamma \cdot (t\gamma + u)} dt = \hat{f}(s\gamma). \end{aligned}$$

Since $\gamma \cdot u = 0$ and $|\gamma| = 1$, we may write $e^{-2\pi i s t} = e^{-2\pi i s \gamma \cdot (t\gamma + u)}$ which implies the third equality. Rotate from γ, e_1, e_2 back to the standard basis to get the final equality and prove the result. □

Nearly all our theorems defined after the Fourier transform have required that any functions used are rapidly decreasing. Hence the significance that the Radon transform preserves the property of being a

given function being rapidly decreasing. The formula in the above theorem will be used in the next section to prove our two central theorems.

6.3. Uniqueness Theorem, Reconstruction Theorem.

Theorem 6.6 (Uniqueness). *If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.*

Proof : The Radon transform is linear due to the linearity of integrals. Therefore, $\mathcal{R}(f) = \mathcal{R}(g) \implies \mathcal{R}(f - g) = 0$. By Lemma 6.5, since $f - g \in \mathcal{S}(\mathbb{R}^3)$, then for any fixed γ , $\mathcal{R}(f - g)(t, \gamma) \in \mathcal{S}(\mathbb{R})$ and $\widehat{\mathcal{R}(f - g)}(s, \gamma) = \widehat{(f - g)}(s\gamma)$. Since the Fourier transform is linear, we have $\widehat{(f - g)}(s\gamma) = \hat{f}(s\gamma) - \hat{g}(s\gamma)$. Then the inverse Fourier transform of both f and g are:

$$(f - g)(s, \gamma) = \mathcal{F}^{-1}[\hat{f}(s\gamma)] - \mathcal{F}^{-1}[\hat{g}(s\gamma)] = \mathcal{F}^{-1}[\mathcal{R}(\hat{f})(s\gamma)] - \mathcal{F}^{-1}[\mathcal{R}(\hat{g})(s\gamma)] = \mathcal{R}(f) - \mathcal{R}(g) = 0.$$

□

Next, we define the aforementioned dual Radon transform.

Definition 6.7. Given F on $\mathbb{R} \times S^2$, we define its **dual Radon transform** by:

$$\mathcal{R}^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma) d\sigma(\gamma).$$

Note the dual Radon transform is NOT the inverse Radon transform. Our domain and range of $\mathcal{R}^* : \mathcal{S}(\mathbb{R} \times S^2) \rightarrow \mathcal{S}(\mathbb{R}^3)$.

We use the dual Radon transform used to prove our last theorem.

Theorem 6.8 (Reconstruction). *If $f \in \mathcal{S}(\mathbb{R}^3)$, then*

$$\Delta(\mathcal{R}^*\mathcal{R}(f)) = -8\pi^2 f,$$

with Δ being the Laplacian operator.

Proof : Given any fixed $\gamma \in S^2$, by Theorem 6.5:

$$\widehat{\mathcal{R}}(f)(s, \gamma) = \hat{f}(s\gamma).$$

Apply Inverse Fourier Transform to both sides:

$$\mathcal{R}(f)(s, \gamma) = \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i t s} ds.$$

Apply Dual Radon Transform to both sides. Recall $x \cdot \gamma = t$:

$$\mathcal{R}^*\mathcal{R}(f)(s, \gamma) = \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i (x \cdot \gamma) s} ds d\sigma(\gamma).$$

Now take the Laplacian. Notice we differentiate with respect to different variables than our variables of integration. Hence the operations are independent and no interchange of operation theorems are required:

$$\begin{aligned} \Delta \mathcal{R}^*\mathcal{R}(f)(s, \gamma) &= \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i (x \cdot \gamma) s} ds d\sigma(\gamma) \\ &= -4\pi^2 \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i (x \cdot \gamma) s} |\gamma|^2 s^2 ds d\sigma(\gamma) \\ &= -8\pi^2 \int_{S^2} \int_0^\infty \hat{f}(s\gamma) e^{2\pi i (x \cdot \gamma) s} s^2 ds d\sigma(\gamma). \end{aligned}$$

Use $\mathcal{P}_{\gamma, t} = \mathcal{P}_{-\gamma, -t}$ to get the final equality. Notice our integral above is defined by a unit vector on the sphere and a radius. This is the same as spherical coordinates in $\mathbb{R}^3$. Therefore, considering s^2 to be the Jacobian from changing variables to spherical coordinates, then this integral is the inverse Fourier transform. Hence, $\Delta \mathcal{R}^*\mathcal{R}(f)(s, \gamma) = -8\pi^2 f(x)$ as desired. □

Thus we have proved our two main results.

REFERENCES

- [1] E. M. Stein and R. Shakarchi, *Princeton Lectures in Analysis I - Fourier Analysis, An Introduction*, Princeton University Press, Princeton, New Jersey, 2003.
- [2] Gerald B. Folland, *Fourier Analysis and its Applications*, Brooks/Cole Publishing Company, Belmont, California, 1992.